quickOrder: An online food ordering and delivery system that provides personalised recommendations and delivery route optimisation

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***Abstract*— In response to the challenges facing the Zimbabwe's food industry where users have to go through generic menus to order their preferred food items and also wait longer times for their deliveries to arrive, I propose this system called quickOrder, which is an AI-powered food ordering and delivery system that utilises advanced machine learning algorithms to offer personalised food recommendations, optimize delivery routes and enhance overall customer satisfaction. By analysing user preferences, past order history and user behaviour, quickOrder will provide tailored suggestions for food items, alleviating the need for manual menu browsing and enhance engagement. Furthermore, the system incorporates real-time route optimisation to ensure accurate delivery estimations and efficient resource allocation, addressing issues of longer waiting times and delayed deliveries. With features like user profiles for storing user preferences and real time order tracking via Google Maps, quickOrder aims to revolutionise the Zimbabwean food delivery sector by providing a seamless, personalised, and efficient ordering and delivery experience to both customers and restaurants.**

Keywords: online food ordering and delivery, machine learning, personalised recommendations, delivery route optimisation.

I. INTRODUCTION

QuickOrder revolutionises the online food ordering and delivery landscape by integrating cutting-edge technology to provide personalised recommendations and optimise delivery routes. Leveraging these state-of the art AI and machine learning algorithms, the system analyses customer preferences, past order history, real-time traffic conditions to offer tailored suggestions for food items [1]. By understanding individual taste of each user, QuickOrder not only enhances user experience but also increases engagement and satisfaction [2].

In addition to personalised recommendations, QuickOrder incorporates advanced delivery route optimisation techniques to ensure efficient and timely deliveries. Through real time analysis of factors such as location, traffic conditions and delivery time windows, the system determines the most optimised routes for delivery routes for delivery drivers, minimising delivery times and costs [3]. This not only benefits customers by reducing waiting

times and ensuring order accuracy but also optimises resource allocation for restaurants leading to improved operational efficiency and cost savings [4].

Moreover, quickOrder offers several unique features that set it apart from the traditional food delivery systems that are currently available in Zimbabwe. Using this system users have the ability to create profiles where they can save their preferences, delivery addresses and payment information, facilitating a seamless and personalised ordering experience [5]. Real time order tracking via Google Maps provides transparency and convenience, allowing customers to monitor the status of their order from preparation to delivery.

By harnessing the power of AI and machine learning, quickOrder not only enhances customer satisfaction and loyalty but also drives business growth for restaurants and delivery partners. With its focus more on personalised recommendations and delivery route optimization, the system is poised to improve the online food ordering and delivery industry, delivering unparalleled convenience and efficiency to stakeholders across the ecosystem.

II. PROBLEM STATEMENT

Some of the primary concerns that are plunging Zimbabwe food delivery industry includes lack of personalised recommendation as customers often has to manually browse through menus to find items of interest, as all the food ordering and delivery system we currently have in Zimbabwe rely on generic menus and do not take into account individual preferences, past order history, location and search history, which can be overwhelming the time consuming leading to customer unsatisfaction and non-engaging experience for customers. As a result of this, restaurants usually miss out on opportunities to upsell and provide personalised recommendations using customer preferences.

In addition, another problem affecting the Zimbabwean food delivery sector is inaccurate delivery estimations and route optimisation as most of the delivery system present in Zimbabwe lacks the capability to optimise delivery routes for drivers effectively. This leads to longer waiting times for customers, delayed deliveries, dissatisfied customers and inefficient resource allocation for restaurant.

To address these issues, this paper proposes an online food ordering and delivery system that implement personalised recommendations and delivery route optimisation to overcome these challenges facing the Zimbabwean food deliver industry and deliver exceptional customer experiences while driving operational efficiency and growth for restaurants and delivery services alike

III. RELATED WORK

Dial and delivery is a popular food ordering and delivery service that allows customers to order food online from various restaurants and have it delivered to their door steps. It is considered as a convenient way for people to enjoy their favourite meals without having to leave the comfort of their homes or offices. This ordering and delivery service is similar to my proposed solution as it allows people to order their desired food items online in the comfort of their homes. As similar as they might be, this current system lacks personalised recommendations as customers have to manually browse through menu to find their preferences which may be quite tiresome and boring reducing customer satisfaction. Therefore my proposed solution is to solve this problem by incorporating recommendation algorithms which

helps in recommending food items of interest to customers thereby increasing customer satisfaction

Munch Zimbabwe is also another system that is already available in Zimbabwe. This again is a popular online food ordering and delivery platform that is currently being used by many food outlets in Zimbabwe. This platform allows customers to order food online from various restaurants and have it delivered to

In the research paper by S.Patil, S.Mahajan, and R.Shinde (2019), they emphasised on the need for a personalised food recommendation system that utilises collaborative filtering and content based filtering algorithms to make suggestions on food items that users may like. The research also focused on the integration of the recommendation algorithms into the online food ordering and delivery platform.. This paper proposes the integration of recommendation algorithms that can provide personalised food recommendations to users based on their historical data, previous preferences and contextual information [6]. This is because these algorithms plays a crucial role in enhancing user experience and engagement in online platforms as the system can help users to discover new dishes, increase user satisfaction, and improve platform retention.

A. Gupta et al. (2019) presented a hybrid approach for personalised food recommendations in online food ordering and delivery platforms. The research was aimed at developing a hybrid approach to personalised food recommendations that combines collaborative filtering and content-based filtering techniques to provide personalised food recommendations as the hybrid approach has shown promising results in improving recommendation accuracy [7]. The hybrid approach is a combination of two or more filtering techniques, and in this study, it's a combination of collaborative filtering techniques and content based filtering techniques [7]. Collaborative filtering techniques uses techniques such as user-based and item-based filtering have been used to analyse user preferences based on their previous orders and recommend similar dishes or restaurants. Content-based filtering techniques utilises user profiles and food item attributes to recommend dishes based on their attributes such as ingredients, cuisine type, or nutritional information [7]

In the paper by S. Sharma et al. (2020), the author appreciated the need for online food ordering and delivery platforms to implement route optimisation algorithms in their system. More so, the author provided an overview of various optimisation techniques used for route planning in food delivery systems. The research paper also discusses different route optimisation algorithms that include ant colony optimisation, genetic algorithms and simulated annealing, which are used to optimise delivery routes and reduce delivery time, thereby increasing the overall system performance and user satisfaction [8].

This research paper by S. Singh et al. (2021), proposes a personalized recommendation system for food delivery services that utilizes machine learning techniques such as collaborative filtering and matrix factorization to provide tailored recommendations based on user preferences, demographics, and past order history [9] With the implementation of machine learning algorithms in the food ordering and delivery industry, food delivery platforms have to incorporate Ai through the use of personalised recommendations algorithms to provide customers with food items or restaurants that they may like, or of similar interest [8]. In this research paper S. Singh et al. (2021), argued that for delivery platform to sustain nowadays there is need to satisfy the customer base, therefore indoor to reduce customer boredom during ordering by going through menus, there is need to provide personalised recommendations to these customers in order to increase their ordering interest due and satisfaction.

This study also highlights the importance of personalized recommendations and delivery route optimization in the food ordering and delivery industry [8]. By leveraging machine learning algorithms and optimization techniques, these systems aim to enhance user experience, increase efficiency, and improve overall customer satisfaction.

IV. SOLUTION

To eradicate the problems of the current food ordering and delivery system that are currently being used in Zimbabwe, I aim develop an online food ordering and delivery system that incorporates machine learning algorithms to provide personalised recommendations to customers and also provide route optimisation, which also have the following objectives:

- To provide customers with personalized recommendations of dishes they may like by analyzing their preferences, past order history, behavior and individual taste.
- To optimize delivery routes by estimating accurate delivery times and allocating delivery resources efficiently.
- To provide customers with real time order tracking through the use of Google maps
- To provide real-time tracking and management of available ingredients, preventing customers from ordering out-of-stock items.

A. Features of the System

The proposed system will have the following features:

1. **Personalized recommendations** -- The system will employ personalized recommendation algorithms in order to provide customers with personalized recommendations for restaurants and food items based on their preferences, previous orders, or popular choices among similar customers. These recommendations to customers are generated in real time and therefore present customers with enhanced ordering experience and encourage exploration of other dishes.
2. **Delivery Route Optimization** -- The system will be incorporated with algorithms for route optimization. In this case, the system will optimize delivery routes based on factors like location of restaurants, customer location, real-time traffic conditions, delivery time windows and the availability of delivery personnel. These algorithms will be used to calculate and determine the most efficient routes in order to minimize travel distance and time, thereby ensuring timely deliveries and optimal resource utilization.
3. **Order Processing** -- The second a customer places an order, the system should process the request promptly, verifies the availability of selected items and also calculate the total cost. The order processing functionality also generates an order confirmation and sends it to the customer and the restaurant for confirmation.
4. **Food Ordering** -- Using this system, users should be able to browse through the available menu, recommendations and restaurants. More so, users should be able to add desired food items to their cart and specify special preferences. In addition, users should also be able to view their order, select delivery or pickup options and proceed to checkout
5. **Order Management** -- Using this system, restaurants should receive order notifications and have a reserved interface to manage and process incoming orders. In addition, the system should support order status updates like "*order received," "preparing," "out for delivery"* and *"delivered."*

6. **Real Time Order Tracking and Notifications** -- Users need to be able to track the status of their orders in real time, including the estimated delivery times and the location of the delivery personnel. Furthermore, customers should receive real time updates on the status of their orders; therefore, notifications have to be sent when the order status is updated.

B. Benefits of the System

The proposed system provided the following benefits:

- For customers:
  1. Personalised recommendations can help customers discover new and exciting dishes that they may not have otherwise known about.
  2. Delivery route optimisation can help ensure that customers food arrives quickly and efficiently, reducing the likelihood of cold or delayed orders.
  3. Overall, a more customized and efficient experience can increase customer satisfaction and loyalty.
- For business:
  1. By offering personalised recommendations, the business can increase customer engagement and sales by introducing customers to new items or encouraging up-selling.
  2. Delivery route optimisation can help the business save on delivery costs by reducing the distance and time required for deliveries.
  3. A better customer experience can lead to more positive reviews, word of mouth referrals, and overall brand loyalty.

C. Solution Architecture

The solution architecture of the system consist of the following:

1. Front end – the web application interface for the admin dashboard and restaurant dashboard is designed using Html, CSS and JavaScript. Then the user mobile applications for the customer and delivery guy are developed using flutter.
2. Backend – the backend for the whole system will be designed using laravel which is a php framework.
3. Recommender system – as for the personalised recommendation part a combination of collaborative and content based filtering algorithms are used to develop a model that provides personalised food recommendations to users based on customer data like past order history, preferences, location and other factors like product ratings, description and reviews.

V. RESULTS AND FUTURE WORKS

A. Results

The system performed well in providing personalised recommendations to customers based on customer data like preferences, past order history, recent searches and customer location and

other factors like product ratings, product description and customer reviews.

TABLE 1

OBJECTIVE ACHIEVED

Objectives	Fully achieved	Partially achieved
To provide customers with personalized recommendations of dishes and restaurants they may like by analyzing their preferences, past order history, behavior and individual taste	✓	
To optimize delivery routes by estimating accurate delivery times and allocating delivery resources efficiently.	✓	
To provide real-time tracking and management of available ingredients, preventing customers from ordering out-of-stock items	✓	
To provide customers with real time order tracking through the use of Google maps	✓	

B. Future Works

Some potential areas of future works for the system could include:

1. Refine the recommendation algorithm so that it incorporates new sources of data like customer demographics, social media profiles and even weather data in order to improve the accuracy of the personalised recommendations that are provided to the customer
2. Integrate the system with a whatsapp chatbots so that customers can be able to order food through whatsapp.
3. Integrate image recognition algorithms so that uses are able to search for food items they like using pictures.

VI. CONCLUSION

The integration of personalized recommendations and delivery route optimization can further enhance the user experience in food ordering and delivery systems. By considering both user preferences and delivery constraints, these systems can provide recommendations that not only align with the user's tastes but also optimize the delivery routes for efficient and timely deliveries. More so, this literature review highlighted the various techniques, approaches, the existing research and developments in the area of food ordering and delivery systems with personalised recommendations and delivery route

optimisation. The studies provide valuable insight into the design, development, and evaluation of effective systems that can provide information on the development of food ordering and delivery systems with personalised recommendations and delivery route optimisation

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